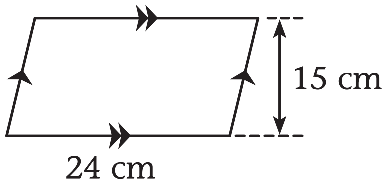
2.5 Using area formulae

There are rules (or formulae) for calculating the area of common plane shapes.

- Area of a **rectangle**
  = length × width = $l \times w$
- Area of a **parallelogram**
  = length × height = $l \times h$
- Area of a **rhombus** or **kite**
  $= \frac{1}{2} \times$ (product of diagonals)
  $= \frac{1}{2} \times X \times Y$
- Area of a **triangle**
  $= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times b \times h$
- Area of a **trapezium**
  $= \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$
  $= \frac{1}{2} \times (a + b) \times h$

EXAMPLE

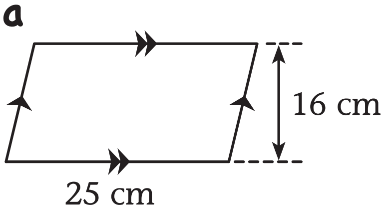
Use the formula $A = l \times h$ to calculate the area of this parallelogram.



Area = 24 × 15
= 360 cm^2

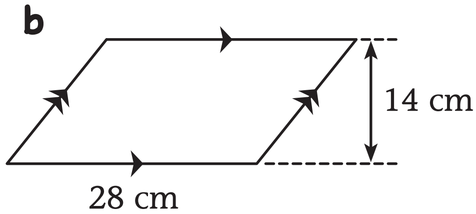
1 Use the formula $A = l \times h$ to calculate the area of each parallelogram.

a



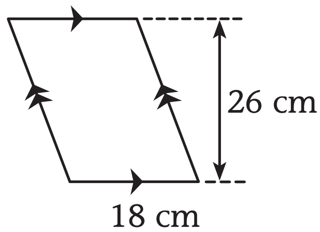
Area = ______________
= __________

b



Area = ______________
= __________

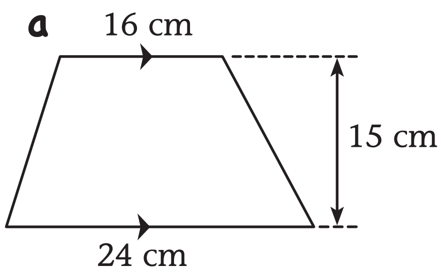
c



Area = ______________
= __________

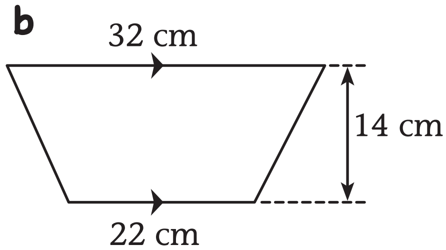
2 Use the formula $A = \frac{1}{2} \times (a + b) \times h$, to calculate the area of each trapezium.

a



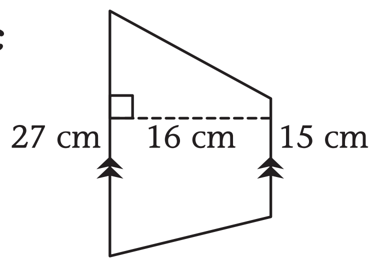
Area = ______________
= __________

b



Area = ______________
= __________

c



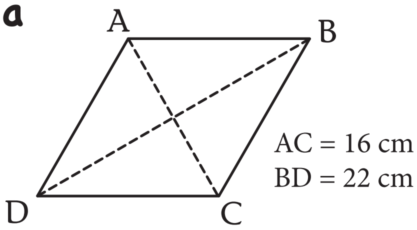
Area = ______________
= __________

2.5 Using area formulae

CONTINUED

3 Use the formula $A = \frac{1}{2} \times X \times Y$ to calculate the area of each rhombus or kite.

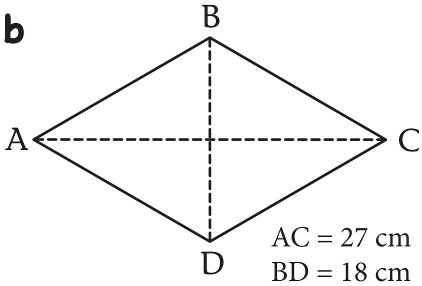
a



Area = ______________

= __________

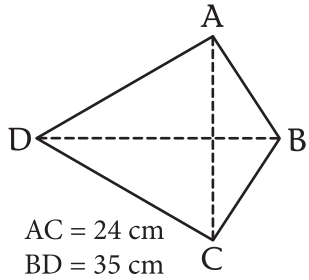
b



Area = ______________

= __________

c



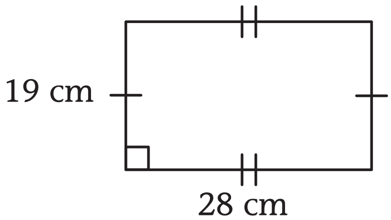
Area = ______________

= __________

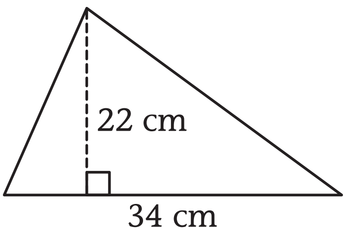
Answer the following questions in your workbook.

4 Select the correct formula and calculate the area of each plane shape.

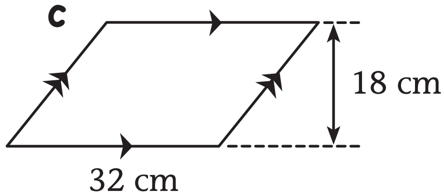
a



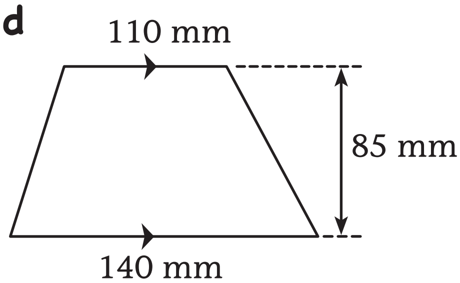
b



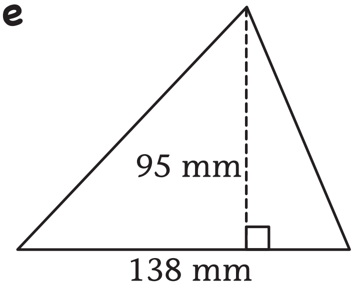
c



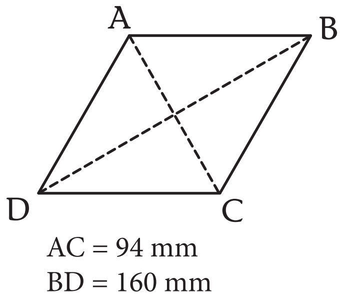
d



e



f



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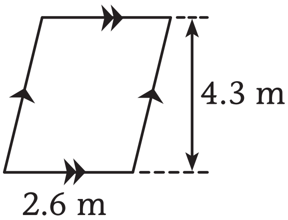
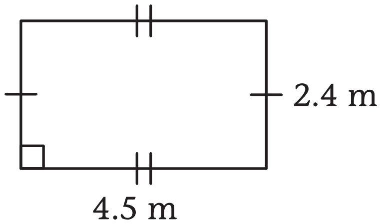


h



i